YUNSUNG LEE

Head of Research, WoRV Team @ MaumAI
Adjunct Professor @ DGIST
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SUMMARY

AI/ML research engineer with strong research foundations (10+ publications in top-tier conferences and over 2,000 citations) and practical experience deploying commercial AI products (over 5M MAUs, Python back-end engineering, ML model deployment). Currently leading research initiatives at MaumAI's WoRV (World Model for Robotics and Vehicle Control) team, specializing in Embodied AI, autonomous agents, and robotics foundation models, while also serving as an Adjunct Professor at DGIST's School of Undergraduate Studies, teaching undergraduate coursework in Physical AI. Expertise spans multimodal AI, autonomous systems, and translating research into deployed products.

WORK EXPERIENCE

Note: Positions at Wrtn Technologies and Riiid served as Alternative Military Service (Technical Research Personnel, 전문연구요원) until April 14, 2025, combining mandatory service with R&D in strategic industries.

MaumAI, Seongnam, South Korea May 2025 - Present
Head of Research, WoRV Team

- Leading research initiatives for WoRV (World Model for Robotics and Vehicle Control), MaumAI's flagship Embodied AI research organization.
- Overseeing the development of foundation models that integrate language, vision, and action for robotics and autonomous driving applications.

DGIST, Daegu, South Korea Aug. 2026 - Present
Adjunct Professor, School of Undergraduate Studies

- Teaching undergraduate coursework in Physical AI at the Daegu Gyeongbuk Institute of Science and Technology (DGIST).

Wrtn Technologies, Seoul, South Korea Oct. 2023 - Apr. 2025
AI Engineer

- Specialized in memory and personalization for autonomous agents and in multimodal capabilities for wrtn. Led core development efforts ranging from ML research and implementation to backend engineering, and delivered the product's key agent features.
- Wrtn reached unicorn status in Aug. 2026 with a valuation above ₩1T.

Riiid (now Socra AI), Seoul, South Korea Apr. 2022 - Oct. 2023
Research Scientist

- Conducted computer vision research for educational AI, building math problem image retrieval for AI:R Math.
- Developed English vocabulary visualizations using text-to-image diffusion models (Santa).

Scatter Lab, Seoul, South Korea Jul. 2021 - Mar. 2022
ML Research Scientist

- Researched a vision-and-language multimodal dialogue system for the chatbot "Luda Lee." (Ref: Make Luda See, Naver Deview 2023.)

CLOVA, NAVER Corp, Seongnam, South Korea Sep. 2020 - Mar. 2021
OCR Team Intern

· Researched self-supervised representation learning, domain generalization, and data augmentations for document images.

HYPERCONNECT, Seoul, South Korea Jul. 2020 - Aug. 2020
ML Team Intern

· Researched adversarially robust semi-supervised learning.

Algorithm Labs, Seoul, South Korea Dec. 2016 - Feb. 2017
Intern

· Developed software and produced algorithm education content.

EDUCATION

Korea University, Seoul, South Korea *Mar. 2019 - Feb. 2022*
M.Sc. Computer Science
Advisors: Seungryong Kim, Jaegul Choo

Carnegie Mellon University, PA, USA *Jan. 2020 - Jul. 2020*
Visiting Scholar, Artificial Intelligence, Language Technologies Institute
Sponsored by IITP under the South Korean government

Hanyang University, Seoul, South Korea *Mar. 2014 - Feb. 2019*
B.Sc. Computer Science

PUBLICATIONS

‡ These publications have accumulated over **2,013 citations** on Google Scholar.

* denotes equal contribution (co-first authors). † denotes co-corresponding authorship.

Suhwan Choi*, Jaeyoon Jung*, Sungkyung Kim, **Yunsung Lee**†, Youngjae Yu†, "PonderPounce: A Pretrained MLLM as an Episode Context Engine for Robot Control," arXiv preprint, 2026 (Under review). arXiv, Project page

Yunsung Lee, Hyeongmin Lee, "Not All Prediction Targets Keep Training-Free Diffusion Guidance on the Manifold," European Conference on Computer Vision (**ECCV'26**), 2026. arXiv, Code

Haebin Seong*, Sungmin Kim*, Yongjun Cho*, Myunchul Joe, Geunwoo Kim, Yubeen Park, Sunhoo Kim, Yoonshik Kim, Suhwan Choi, Jaeyoon Jung, Jiyong Youn, Jinmyung Kwak, Sunghee Ahn, Jaemin Lee, Younggil Do, Seungyeop Yi, Woojin Cheong, Minhyeok Oh, Minchan Kim, Seongjae Kang, Samwoo Seong, Youngjae Yu†, **Yunsung Lee**†, "CostNav: A Navigation Benchmark for Real-World Economic-Cost Evaluation of Physical AI Agents," The First Workshop on Scalable Robot Learning Systems at CVPR 2026 (**ScaleBot @ CVPR'26**), 2026. arXiv, Project page

Jisu Nam, Yicong Hong, Chun-Hao Huang, Feng Liu, JoungBin Lee, Jiyoung Kim, Siyoon Jin, **Yunsung Lee**, Jaeyoon Jung, Suhwan Choi, Seungryong Kim†, Yang Zhou†, "WorldCam: Interactive Autoregressive 3D Gaming Worlds with Camera Pose as a Unifying Geometric Representation," The CVPR 2026 Workshop on Video World Models (**VideoWorldModel @ CVPR'26**), 2026. arXiv, Project page

Suhwan Choi, **Yunsung Lee**, Yubeen Park, Chris Dongjoo Kim, Ranjay Krishna, Dieter Fox, Youngjae Yu, "vla-eval: A Unified Evaluation Harness for Vision-Language-Action Models," The ICRA 2026

Workshop on From Data to Decisions (**From Data to Decisions @ ICRA'26**), 2026. arXiv, Project page

Suhwan Choi*, Jaeyoon Jung*, Haebin Seong*, Minchan Kim, Minyeong Kim, Yongjun Cho, Yoonshik Kim, Yubeen Park, Youngjae Yu†, **Yunsung Lee**†, "D2E: Scaling Vision-Action Pretraining on Desktop Data for Transfer to Embodied AI," The International Conference on Learning Representations (**ICLR'26**), 2026. arXiv, Project page

Yohan Lee*, Sungho Park*, Sangwoo Han*, **Yunsung Lee***†, Yongwoo Song, Adam Lee, Jiwung Hyun, Jaemin Kim, Seungtaek Choi, HyeJin Gong†, "SAFARI: Sample-specific Assessment Framework for AI in Real-world Interactions," Findings of the Annual Conference of the North American Chapter of the Association for Computational Linguistics (**Findings of NAACL'25**), 2025 (Accepted, but withdrawn due to corporate policy).

Jin-Young Kim*, Soonwoo Kwon*, Hyojun Go*, **Yunsung Lee**, and Seungtaek Choi, "ScoreCL: Augmentation-Adaptive Contrastive Learning via Score-Matching Function," Machine Learning (Springer Journal), 2025. arXiv

Yunsung Lee*, Jin-Young Kim*, Hyojun Go*, Myeongho Jeong, Shinhyeok Oh, and Seungtaek Choi, "Multi-Architecture Multi-Expert Diffusion Models," Annual AAAI Conference on Artificial Intelligence (**AAAI'24**), 2024. arXiv

Hyojun Go*, Jin-Young Kim*, **Yunsung Lee***, Seunghyun Lee, Shinhyeok Oh, Hyeongdon Moon, and Seungtaek Choi, "Addressing Negative Transfer in Diffusion Models," Conference on Neural Information Processing Systems (**NeurIPS'23**), 2023. arXiv, Code

Shinhyeok Oh*, Hyojun Go*, Hyeongdon Moon, **Yunsung Lee**, Myeongho Jeong, Hyun Seung Lee, and Seungtaek Choi, "Evaluation of Question Generation Needs More References," Findings of the Annual Meeting of the Association for Computational Linguistics (**Findings of ACL'23**), 2023. arXiv

Hyun Seung Lee*, Seungtaek Choi*, **Yunsung Lee**, Hyeongdon Moon, Shinhyeok Oh, Myeongho Jeong, Hyojun Go, and Christian Wallraven, "Cross Encoding as Augmentation: Towards Effective Educational Text Classification," Findings of the Annual Meeting of the Association for Computational Linguistics (**Findings of ACL'23**), 2023. arXiv

Hyojun Go*, **Yunsung Lee***, Jin-Young Kim*, Seunghyun Lee, Myeongho Jeong, Hyun Seung Lee, and Seungtaek Choi, "Towards Practical Plug-and-Play Diffusion Models," The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2023 (**CVPR'23**), 2023. arXiv, Code

Yunsung Lee*, Gyuseong Lee*, Kwangrok Ryoo*, Hyojun Go*, Jihye Park*, Seungryong Kim, "Towards Flexible Inductive Bias via Progressive Reparameterization Scheduling," ECCV 2022 Visual Inductive Priors for Data-Efficient Deep Learning Workshop (**ECCVW'22**), 2022. arXiv

Seokju Cho*, Sunghwan Hong*, Sangryul Jeon, **Yunsung Lee**, Kwanghoon Sohn, and Seungryong Kim, "CATs: Cost Aggregation Transformers for Visual Correspondence," Conference on Neural Information Processing Systems (**NeurIPS'21**), 2021. arXiv, Code

Junbum Cha, Hancheol Cho, Kyungjae Lee, Seunghyun Park, **Yunsung Lee**, and Sungrae Park. "SWAD: Domain Generalization by Seeking Flat Minima," Conference on Neural Information Processing Systems (**NeurIPS'21**), 2021. arXiv, Code

Yunsung Lee, Teakgyu Hong, Han-Cheol Cho, Junbum Cha, and Seungryong Kim. "HoughCL: Finding Better Positive Pairs in Dense Self-supervised Learning," ICML 2021 Workshop: Self-Supervised Learning for Reasoning and Perception (**ICMLW'21**), 2021. arXiv

Yunsung Lee, Teakgyu Hong, and Seungryong Kim. "Data Augmentations for Document Images," The AAAI-21 Workshop on Scientific Document Understanding (**AAAIW'21**), 2021.

Seokeon Choi, Junhyun Lee, **Yunsung Lee**, and Alexander Hauptmann. "Robust Long-Term Object Tracking via Improved Discriminative Model Prediction," The ECCV-20 Workshop on Visual Object Tracking Challenge (**ECCVW'20**), 2020. arXiv, Code

Junsoo Lee*, Eungyeup Kim*, **Yunsung Lee**, Dongjun Kim, Jaehyuk Chang, and Jaegul Choo, "Reference-Based Sketch Image Colorization using Augmented-Self Reference and Dense Semantic Correspondence," IEEE Conference on Computer Vision and Pattern Recognition (**CVPR'20**), 2020. arXiv, Project page

Sookyung Kim*, Sunghyun Park*, Sunghyo Chung*, Joonseok Lee, **Yunsung Lee**, Hyojin Kim, Mr Prabhat, and Jaegul Choo, "Learning to Focus and Track Extreme Climate Events," British Machine Vision Conference (**BMVC'19**), 2019. Accepted as Spotlight Presentation (6.9% acceptance rate for spotlight papers).

TECHNICAL STRENGTHS

ML Research	Vision-language multimodality, Diffusion Models, Autonomous Agents
Tools	Python, PyTorch, Git, Docker, FastAPI, Elasticsearch

HONORS & AWARDS

ICML 2026 Gold Reviewer — Recognized among the top reviewers based on area-chair ratings of submitted reviews; awarded complimentary registration, 2026

5th place, RLT-DiMP team, VOT-Long-Term, ECCVW Visual Object Tracking Challenge, 2020

4th place, HYU Programming Contest (Advanced Division), Seoul, Korea, 2015

Dean's citation for ACM-ICPC Daejeon Regional, Hanyang University, Seoul, Korea, 2014

OTHER ACTIVITIES

Invited Speaker Aug. 2026
Telecommunications Technology Association (TTA), Seongnam, South Korea

- "World Models and World-Action Models: From VLAs to WAMs." Physical AI seminar. [slides]

Invited Mentor Aug. 2026
Korean Conference on Computer Vision (KCCV 2026), Busan, South Korea

- Mentoring session "Work & Life." [program]

Invited Speaker May 2026
Hankuk University of Foreign Studies (HUFS), Division of Language & AI

- "Robotics Foundation Models: Trajectories and open problems." Multimodal AI course. [slides]

Invited Speaker Apr. 2026
State University of New York (SUNY) Korea

- Invited talk for STEM undergraduate students.

Judge Nov. 2025
Hack@thon

- Served as a judge for the Hack@thon sponsored by Lovable at Sogang University. [link]

Conference Reviewer

- The IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**) (2023 -)

- The IEEE/CVF International Conference on Computer Vision (**ICCV**) (2023 -)
- Advances in Neural Information Processing Systems (**NeurIPS**) (2024 -)
- European Conference on Computer Vision (**ECCV**) (2024 -)
- The International Conference on Learning Representations (**ICLR**) (2024 -)
- The International Conference on Machine Learning (**ICML**) (2026 -) — Gold Reviewer Award, 2026
- ACL Rolling Review for **ACL, EMNLP** (2024 -)

Selected Open-source Contributions

- Core contributing member of the open-world-agents organization, which builds tools for open-world agent R&D. Contributions include open-world-agents and desktop-env, a real-time, high-frequency, real-world desktop environment for desktop-based ML development (agents, world models, etc.).
- Contributor to mem0, a memory layer for AI apps.
- Creator of awesome-visual-representation-learning-with-transformers, a curated "awesome list" (200+ stars) started in 2021, when Vision Transformer research was just taking off.

PR12, TensorFlow Korea, South Korea Jun. 2020 - Present
Member

- Active member of TensorFlow Korea, Korea's largest machine learning research community, participating in an advanced study group on recent ML research.

TEDxHanyangU Sep. 2017 - Jun. 2018
Organizer

- Experience Catalyst, 2018
- Web Engineer, 2017

SW Maestro, Ministry of Science and ICT, Seoul, South Korea Aug. 2017 - Jan. 2018
Trainee

- Government-run software talent development program under Korea's Ministry of Science and ICT.
- Completed a CPA document automation project and learned the fundamentals of computer vision and machine learning, which prompted the decision to pursue graduate study.

ALOHA (Algorithm research team of Hanyang University) Nov. 2015 - Oct. 2016
Team Leader

- Taught algorithms to team members. Hosted several programming contests.
- Test writer & presenter, The First KSH (Korea, Sookmyung W, Hanyang Univ.) Algorithm Camp.
- Chief test writer, HYU Programming Contest.

Home Bartender
Licensed Bartender

- Certified under Korea's national Craftsman Bartender license — a hobby that rewards both precision and improvisation.